Retail Sales & Profitability Analysis

Tools: Python | SQL (MySQL) | Power BI

1. Business Objective

The objective of this project is to analyse retail transaction data to understand sales performance, identify revenue and profit drivers, and highlight loss-making areas to support data-driven business decisions.

2. Dataset Overview

The dataset contains retail transaction records including product details, quantities sold, pricing, discounts, sales dates, and profit information. The data represents sales across multiple product categories and regions.

3. Data Preparation

- Handled missing and inconsistent values in key fields.
- Converted date fields into appropriate formats for time-based analysis.
- Created derived metrics such as total sales and profit where required.

4. Analysis Performed

- Sales and revenue analysis across products, categories, and regions.
- Profitability analysis to identify high-margin and loss-making products.
- Trend analysis to understand monthly and overall sales performance.
- Evaluation of discounts and their impact on profit.

5. Key Insights

- A small number of products and categories contributed a major share of total revenue.
- Certain products generated high sales volumes but low or negative profit margins.
- Sales showed clear seasonal and monthly trends across regions.
- Discounts did not always result in proportional increases in profitability.

6. Dashboard & Visualization

A one-page Power BI dashboard was created to present key KPIs, sales trends, product performance, and profitability metrics in an interactive and business-friendly format.

7. Conclusion

This project demonstrates an end-to-end retail analytics workflow, combining Python-based data preparation, SQL-driven analysis, and Power BI visualization to deliver actionable business insights.